1. Calculate:

a) 4.8 + 5.7 _____ b) 7 x 6 _____

c) 56 ÷ 8 _____ d) 112 – 39 _____

e) 5/6 + 4/12 _____

---

2. This number chain is made using the formula

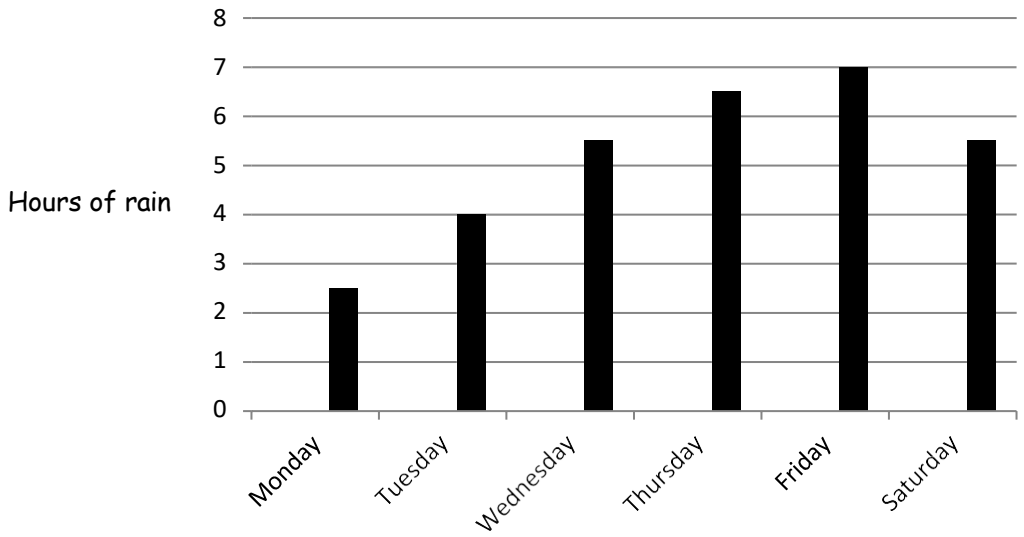
double each number and subtract three to make the next number.

7 ⟷ 11 ⟷ 19 ⟷ ? ⟷ 67

What is the missing number? _____

---

3. This bar graph shows the number of hours it rained for six days.



How many more hours of rainfall were there on Friday than on Monday?

_____

---

4. On the new HS 1 line trains travel at speeds of 192 km per hour. How many km can they travel in 25 minutes? _____

---

5. A 2.5 litre bottle of fizzy drink is divided equally between 8 people. How many millilitres will each person receive? _____

6. The sum of three consecutive odd numbers is 27. What is the largest number of the three? _____

---

7. David, Peter and Sue were given these marks for their English tests.

David	31 out of 50
Peter	36 out of 50
Sue	23 out of 50

a/ What was Sue's percentage for English? _____

b/ What is the average of their marks for English? _____

c/ What was the difference in the percentage scores of David and Peter? _____

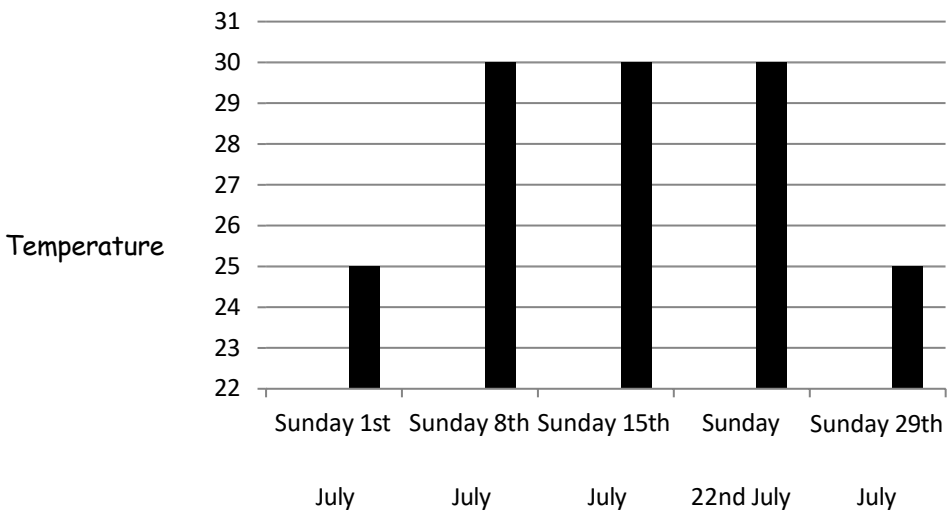
---

8. Find the missing number in each of these calculations.

a/ 16 = 7 + 2 + ? _____

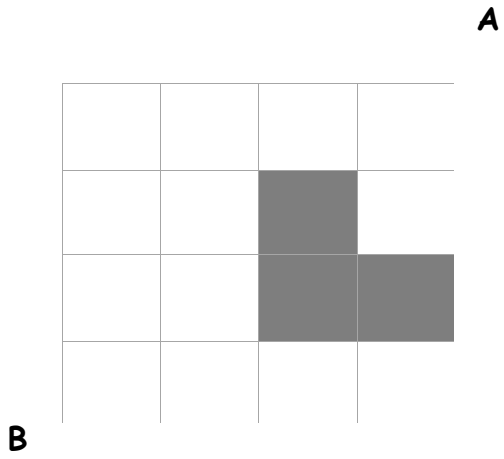
b/ -8 + 5 = ? _____

9. The bar chart shows the maximum temperature for each of 5 Sundays in July in Athens.

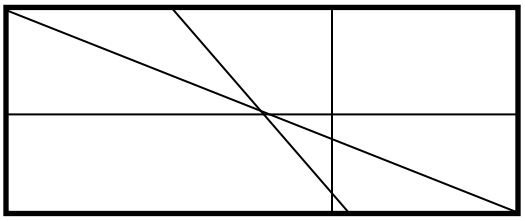


What was the average maximum temperature for these 5 Sundays? _____

10. A square is divided into 16 equal smaller squares. Three of these are already shaded. Shade in the least number of squares necessary to make the diagonal line joining corners A and B a line of symmetry.



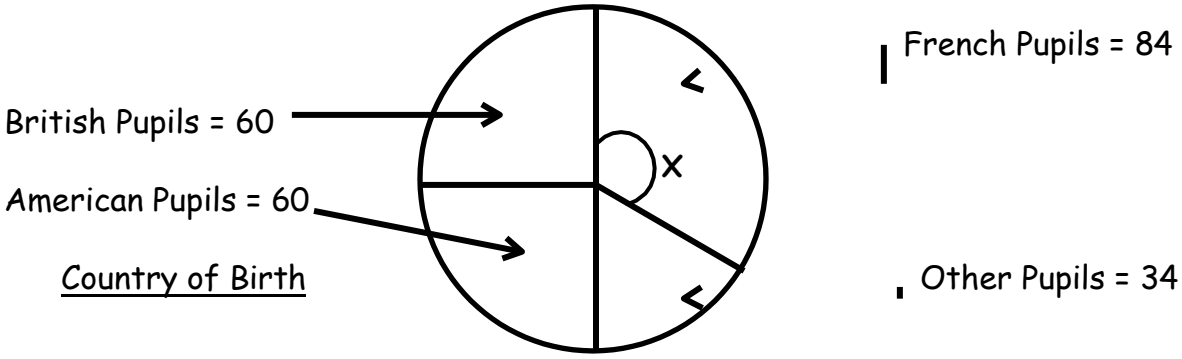
11. How many triangles of any size are there in this diagram? _____



12. At the delicious Cookery School there are 100 people including all students and all staff. Sixty students study during the day and twenty five students study in the evening. Of these, 8 students study both during the day and in the evening, how many staff are there at the school? _____

13. At the English School in Istanbul each pupil makes one visit a year to their country of birth. 84 children were born in France. There are 240 children at the school.

Calculate the size of angle x. _____



14. Calculate: **a/** 7.63 – 5.17 _____

b/ 9.4 x 100 _____

c/ 268 x 3 _____

d/ 2 2/6 + 4 2/3 _____

---

15. The bakery charges **x** pounds for a chocolate cake and **y** pence for a lemon cake.

a/ Write down the expression using **x** and **y** for the cost of one chocolate cake and two lemon cakes. _____

b/ John bought **6** of these lemon cakes for **£3.42**, use this information to work out what **y** stands for. _____

16. Air is made up from oxygen and nitrogen. In the air the ratio of oxygen to nitrogen is 1:4

How much oxygen is there in 30 litres of air? _____

---

17. **6** **7** **26** **24** **25**

Which of these numbers is:

a/ a prime number? _____

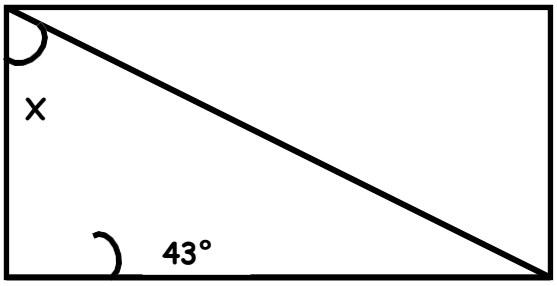
b/ a multiple of 8? _____

c/ a factor of 30? _____

d/ a square number? _____

---

18. What is the size of the angle marked X in the rectangle? _____



---

19. John works out that the average of 7 of his end of term exams is 64%. His eighth and last exam result is 94%.

How much will John's average increase by when he adds in his last result?

_____

---

20. What is the next number in the sequence? _____

173 85 41 19 8

---

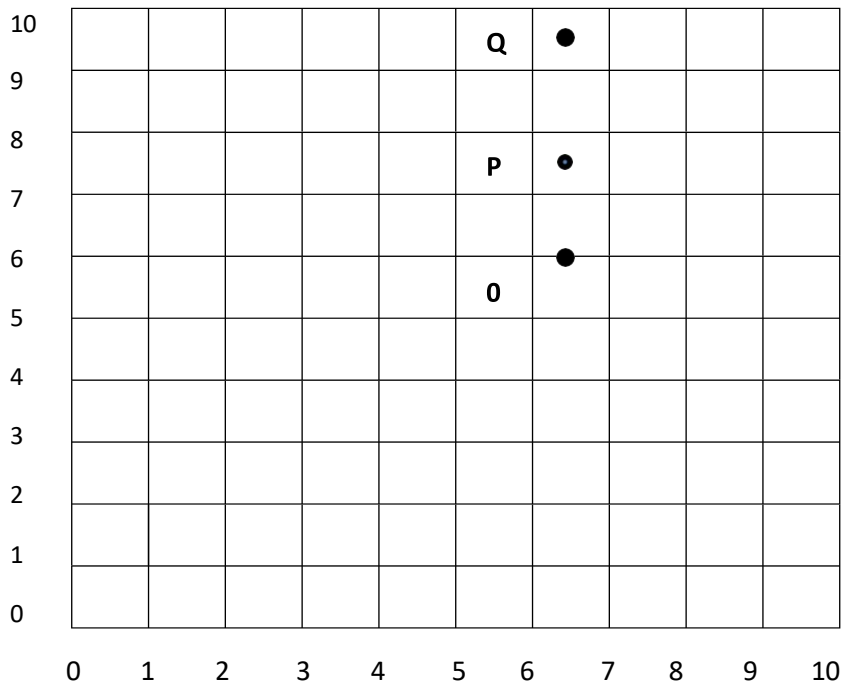
21. The average number of corners awarded to the British football team over their last 5 matches was 4.4. If they received a total of 16 corners in 4 of these matches, how many were they awarded in the fifth match? _____

---

22. Here is part of a multiplication table. Fill in the missing numbers.

60	72	84
65		91
70		98

23.



The line OQ is rotated clockwise through 90°.

What are the new coordinates of P? _____

---

24. An orangutan takes 15 seconds to open a jar and can open 2 jars at the same time. How many jars can the orangutan open in an hour? _____

---

25. Which of these numbers is closest to zero? _____

a. 1.04 **b.** 0.004 **c.** 0.3 **d.** 0.051 **e.** 0.103

END OF TEST

If you have time

- Go back and complete any questions you missed out.

- Check your answers carefully.

CORE SKILLS 9

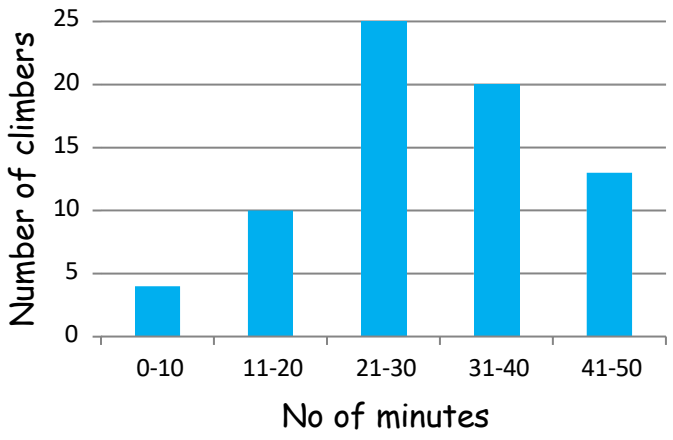
1

Every weekday the chickens eat corn. At the weekends they eat pellets. The farmer uses one 20kg bag of corn on each weekday; this costs £0.37 per kg.

How much is his corn bill for the week?

2

The rock climbing club recorded the time it took members to climb the rock wall.



How many climbers took more than 30 minutes to climb the wall?

3

A flower bed is 156cm long. Choose the closest measure of this length, in feet and inches, from the options provided.

5ft 11inches 6ft 4inches 4ft 8inches 5ft 8inches 4ft 10inches

4

A large chocolate coated biscuit weighs 36 grams. Only 40 % of its weight is chocolate.

How many grams of chocolate are in a packet of 30 biscuits?

400g 360g 512g 432g 430.2g

5

A shelf can store twelve rows of twelve books. How many shelves would be needed for 1080 books?

6

Alfie is playing darts. In darts you need to score 501 to win the leg. You have 3 darts per turn and must finish on a double. On his first three turns Alfie scores 123, 180 and 150. On his last turn if Alfie throws a 10 and an 8, what does he need with his final dart in order to win the leg?

CORE SKILLS 9

7

Sophie trained every Monday and Thursday for 6 weeks before her swimming competition. She recorded the number of lengths of the pool she swam each week. The mean number of lengths was 65. How many lengths did she swim in week 5?

Week	Lengths
1	57
2	63
3	78
4	89
5	
6	44

8

What is the lowest number made by using 2, 13 and 7 as factors?

9

A number is made up of 8 units, 2 tens and 7 hundreds. Peter divides this number by 10.

What number does Peter get?

7820 72.8 78.2 87.2 27.8

10

Jack recorded the number of baskets scored for the first eleven games of the basketball season, as follows:

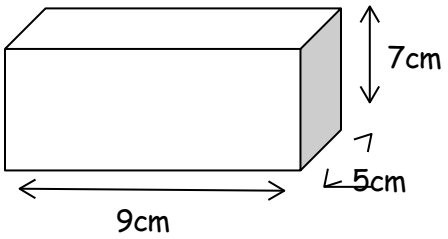
23 31 8 42 38 13 2 5 48 51 19

What was the median score for the team?

CORE SKILLS 10

1

Look at the cuboid below. Its dimensions are 7cm by 5cm by 9cm.



What is the total length of all the edges of the cuboid?

186cm 72cm 84cm 42cm 168cm

2

From the cuboid in question 1, what is the area of the largest face?

$72cm^2$ $54cm^2$ $45cm^2$ $81cm^2$ $63cm^2$

3

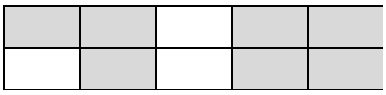
Mrs Taylor won £980 in a competition. She divided the money between her four children – Brian, Simon, Sarah and Molly. Simon then gave a quarter of his money to his best friend Rupert. How much money did Rupert receive?

4

What is the highest number that can be divided equally into both 178 and 159?

5

The rectangle below is divided into squares of equal area.



What percentage of the rectangle is shaded?

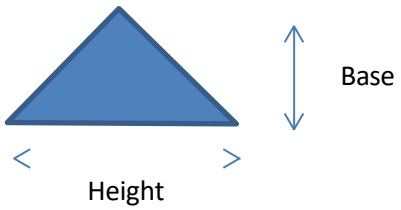
30% 70% 7% 80% 3%

6

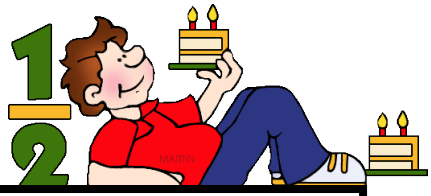
Look at this rule. Triple the previous number and add one.
Complete the sequence:

16 _____ _____ _____

CORE SKILLS 10

7	Four of the following are correct. Select the incorrect sum from the options provided: (42/6) + 9 = 16 7 = (42/6) 42 = (16 - 9) x 6 42 - 16 = 9 x 6 42/6 – 7 = 0
8	13^2 =
9	One of the equations below is the area of a triangle. Please select the correct one.  ($\frac{1}{2}$ base) + Height 2 x (Height - Base) $\frac{1}{2}$ (base + height) 2 x (base x height) $\frac{1}{2}$ (base x height)
10	In women's tennis there are 4 points in a game, 6 games in a set and 2 sets in a match. What is the minimum possible number of points needed to win a match?

Maths Reasoning

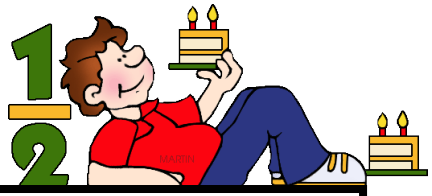


These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

Estimate how many people are in this crowd.

MR 25

Maths Reasoning



These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

1 2 3 4 5 6 7 8 9

Choose four different digits from those given above.

Draw a four box square like the one below.

Put a single digit in each box.

This makes two 2-digit numbers by reading across and two 2-digit numbers by reading down.

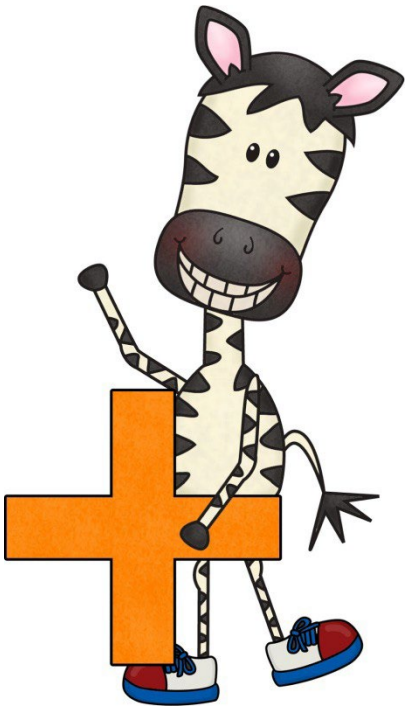
Add up all four of the 2-digit numbers.

For example if the total is 200 then the following solution is possible.

1	9
7	2

19 + 72 + 17 + 92 = 200

How many different ways of making 100 can you find?



MR 26

Maths

Maths Type 2 Paper 6

Page 1

1.

a) 10.5

b) 42

c) 7

d) 73

e) $1\frac{1}{6}$

2. 35

3. 4.5hr

4. 80km

5. 312.5ml

Page 2

6. 11

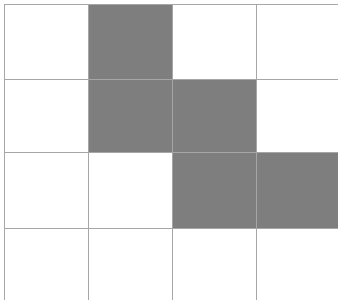
7. a. 46% b. 30 c. 10%

8. a. 7 b. -3

9. 28°C

Page 3

10.



11. 20

12. 23

13. 126°

Page 4

14. a. 2.46 a. 940 b. 804 d. 7

15. a. 2y + x b. 57p

1. 6 litres

2. a. 7 b. 24 c. 6 d. 25

Page 5

3. 47°

4. 3.75%

5. 2.5

6. 6

7. 78, 84

Page 6

8. (8, 6)

9. 480

10. 0.004

Core Skills 9

1. £37.00
2. 33
3. 5ft 11inches
4. 432g
5. 8 shelves
6. Double 15
7. 59
8. 182
9. 72.8
10. 23

Core Skills 10

1. 84cm
2. 63cm^2
3. £61.25
4. 28302
5. 70%
6. 16, 49, 148, 445
7. 42 - 16 = 9 x 6
8. 169

9. $\frac{1}{2}$ (base x height)
10. 48 points

Maths Reasoning 25

There is no precise answer, but by using logic you can achieve a good estimate.

Example:

Take a small area such as a 1-inch square. Estimate numbers within it, and then estimate how many of the small areas make up this larger area. Multiply them together to calculate your overall estimate.

Maths Reasoning 26

Two solutions are possible and since swapping the bottom left and top right numbers does not affect the equation, an additional two solutions are created.

So there are FOUR solutions:

2	1
3	8

2	3
1	8

1	2
4	7

1	4
2	7